Stable AC-personal

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1 Introduction

This note fixes the definitions behind every search method in AC-SolverX and reports the score of each method on a common benchmark. Each score is the number of presentations of a fixed set that the method certifies within a stated budget. Each certificate is a sequence of moves, replayed independently on the input presentation before it is counted.

Notation. Let $X = \{x_1, \dots, x_n\}$ and let $F(X)$ be the free group on X. A word over $X^{\pm 1}$ is *freely reduced* if it contains no factor aa^{-1} with $a \in X^{\pm 1}$, and *cyclically reduced* if every cyclic permutation of it is freely reduced. For $w \in F(X)$, $|w|$ denotes the length of its freely reduced form. For two generators we write x, y. In tables, an upper-case letter denotes an inverse: `xxxYYYY` is x^3y^{-4}.

Definition 1.1 (Balanced presentation)**.** A *presentation* is a pair $P = \langle X \mid R \rangle$ with $R = (r_1, \dots, r_m)$ a tuple of cyclically reduced words. It presents the group $G_P = F(X)/\langle\langle R \rangle\rangle$, where $\langle\langle R \rangle\rangle$ is the normal closure of R in $F(X)$. P is *balanced* if $m = n$, and n is its *rank*. The *total length* of P is $\lambda(P) = \sum_i |r_i|$. The *trivial presentation* of rank n is $T_n = \langle x_1, \dots, x_n \mid x_1, \dots, x_n \rangle$.

Benchmarks. For $n \geq 1$ and $w \in F(x, y)$ with exponent sum 0 in x, the Miller–Schupp presentation

$$\mathrm{MS}(n, w) = \langle x, y \mid x^{-1}y^n x\, y^{-(n+1)},\ x^{-1}w \rangle$$

presents the trivial group.

- **MS640** is the set of the 640 presentations of the 1,190-row Miller–Schupp benchmark that the LENGTH greedy of Section 1.1.2 solves within 10^6 pops at cap 24. Every presentation in MS640 is therefore AC-trivial, and a method that fails on a row has failed on a solvable instance.
- **bench60** is a 60-row subset of MS640 spanning its difficulty bins.
- **ladder200** contains 188 presentations graded into ten difficulty levels by the node count of the LENGTH greedy on record (144 Aut-minimal AC19 orbits and 44 MS640 rows).
- **AC19** is the set of 72,779 representatives of the $\mathrm{Aut}(F_2)$-orbits of the 156,762 presentations of the extended AC19 data set.

Units. A *pop* (or *node*) is one removal from the search heap. A budget of B pops is per presentation, and a failed search is charged all B pops. Path lengths are counted in S-moves (Definition 1.5) unless stated otherwise. Scores measured in different units (pops, the hybrid's work units) are not compared with each other.

1.1 AC conjecture

Definition 1.2 (AC moves)**.** Let $P = \langle X \mid r_1, \dots, r_n \rangle$ be balanced. An *AC move* replaces a single relator r_i and leaves the others unchanged:

$$\begin{aligned} &\text{(AC1)} \quad r_i \mapsto r_i^{-1}; \\ &\text{(AC2)} \quad r_i \mapsto r_i\, r_j, \qquad j \neq i; \\ &\text{(AC3)} \quad r_i \mapsto u\, r_i\, u^{-1}, \qquad u \in F(X). \end{aligned}$$

After each move the new relator is freely and cyclically reduced.

Each move has an inverse composed of moves of the same kinds: $r_i \mapsto r_i r_j^{-1}$ is AC1 on r_j, then AC2, then AC1 on r_j. A transposition of two relators is also a composite of AC1 and AC2. Hence the relation $P \sim_{AC} Q$, meaning that Q is obtained from P by a finite sequence of AC moves, is an equivalence relation. P is *AC-trivial* if $P \sim_{AC} T_n$.

Lemma 1.3 (Invariants)**.** *Let $P \sim_{AC} Q$ be balanced presentations on X.*

1. *$\langle\langle R_P \rangle\rangle = \langle\langle R_Q \rangle\rangle$; in particular $G_P \cong G_Q$.*
2. *Let $M(P) \in \mathbb{Z}^{n \times n}$ be the exponent-sum matrix, where M_{ij} is the exponent sum of x_j in r_i. Then $|\det M(P)| = |\det M(Q)|$.*

Consequently, an AC-trivial presentation presents the trivial group and satisfies $|\det M| = 1$.

Proof. (1) Every move replaces r_i by an element of the normal closure of the current relators, and every move is invertible. (2) Abelianization maps r_i to row i of M. AC1 negates a row, AC2 adds row j to row i, and AC3 fixes every row. Each therefore multiplies M on the left by a unimodular matrix. Finally, $M(T_n) = I_n$. □

Neither invariant is sufficient for AC-triviality. Every verifier in the code base checks $|\det M| = 1$ independently of the solver.

Conjecture 1.4 (Andrews–Curtis, 1965)**.** *Every balanced presentation of the trivial group is AC-trivial.*

The conjecture is open and is generally believed to be false. The standard family of potential counterexamples is the Akbulut–Kirby family

$$\mathrm{AK}(n) = \langle x, y \mid x^n y^{-(n+1)},\ xyx\, y^{-1} x^{-1} y^{-1} \rangle, \qquad n \geq 2.$$

AK(2) is AC-trivial, and AK(3) is the standard test case.

1.1.1 Substitution

Definition 1.5 (Substitution move)**.** For a cyclically reduced word r and $0 \leq k < |r|$, let $\mathrm{rot}_k(r)$ be r with its last k letters moved to the front. On a balanced presentation of rank n, a *substitution move* (S-move) is a tuple (i, j, s, k_1, k_2) with $i \neq j$ and $s = \pm 1$. It acts by

$$r_i \ \longmapsto\ \mathrm{rot}_{k_1}(r_i) \cdot \mathrm{rot}_{k_2}\big(r_j^{\,s}\big),$$

followed by free and cyclic reduction. The move is admissible only if the last letter of $\mathrm{rot}_{k_1}(r_i)$ is the inverse of the first letter of $\mathrm{rot}_{k_2}(r_j^{\,s})$, that is, if the product cancels at the seam.

Lemma 1.6. *Every S-move is a composite of AC moves.*

Proof. $\mathrm{rot}_k(r) = c\,r\,c^{-1}$ for a suffix c of r, so each rotation is an AC3 move. Replacing r_j by r_j^{-1} is AC1, and the product is AC2. Undoing the conjugation and the inversion of r_j restores the other relator. □

Example 1.7. On AK(3) = $\langle x, y \mid \texttt{xxxYYYY}, \texttt{xyxYXY}\rangle$, the move $(1, 2, +1, 0, 5)$ forms the product `xxxYYYY` · `yxYXYx`, which reduces to `xxxYYYxYXYx`. The child has total length 17. AK(3) has 42 admissible S-moves and 38 distinct children.

Definition 1.8 (State)**.** A *state* is a balanced presentation modulo cyclic rotation of each relator, inversion of each relator and permutation of the relators. It is represented by a canonical form: each relator is replaced by the least rotation of itself or its inverse, and the tuple of relators is sorted.

1.1.2 Greedy Substitution

Definition 1.9 (Greedy best-first search)**.** The search is parametrised by:

- a priority π on states;
- a budget $B \in \mathbb{N}$;
- a relator cap $c \in \mathbb{N}$.

The search maintains a heap of states ordered lexicographically by $(\pi(Q), \mathrm{depth}(Q), \kappa(Q))$, where κ is an injective key. This order is total, so the run is deterministic. The search also maintains the set of states already discovered. Starting from the input, it performs at most B pops. A popped state is *solved* if every relator has length 1. Otherwise the search pushes each child under an admissible S-move that is undiscovered and has every relator of length at most c. The certificate is the chain of parent moves.

The *LENGTH* greedy is the case $\pi = \lambda$.

A solved state with all relators of length 1 satisfies $|\det M| = 1$ by Lemma 1.3, so it is a signed permutation of T_n.

Remark 1.10 (Budget and cap).

1. A search with budget B performs exactly the first B pops of the same search with any budget $B' > B$. Hence the set $\mathcal{S}_B$ of solved presentations is non-decreasing in B.
2. Suppose that no child discarded by the cap has a heap key at most that of a state popped later. Then the run is identical under every cap $c' \geq c$. Every run records the counters that decide this condition.

Score of the LENGTH greedy. All MS640 runs use cap $c = 24$.

budget B (pops)	$\lvert\mathcal{S}_B\rvert$ (of 640)	mean pops (solved rows)	median pops (solved rows)	mean path length (S-moves)
100	455			
1,000	554	65.4	9	12.09
10,000	606	389.5	10	19.97
50,000	628			
10^6	640	4,963.0	11	36.77

The cost distribution at $B = 10^6$ is strongly concentrated (maximum 574,959 pops). The 529 rows in difficulty bins 0–3 account for 0.69% of the 3,176,297 pops, the 6 most expensive rows for 66.8%, and the 20 most expensive rows for 86.2%. The cost is not an invariant of the $\mathrm{Aut}(F_2)$ orbit: a single orbit of MS640 ranges from 14,415 to 272,953 pops. On the other benchmarks at $B = 10^4$, the LENGTH greedy solves:

- bench60: 40/60 (and 29/60 at $B = 10^3$);
- ladder200: 40/188;
- AC19: 71,936/72,779, at cap 48.

1.1.3 S20_MK2 Substitution

Definition 1.11 (Block features)**.** Read each relator as a cyclic word. A *block* is a maximal cyclic run of letters from one generator, with x and x^{-1} counted as the same generator.

- The *knot number* of a relator is 0 if it is a pure power, and otherwise the larger of its numbers of x-blocks and y-blocks. $\mathrm{MK}(Q)$ is the larger of the knot numbers of the two relators of Q.
- Let $\bar{s}_x$ and $\bar{s}_y$ be the mean lengths of the x-blocks and of the y-blocks, pooled over both relators. $S(Q)$ is the smaller of the non-zero values among $\bar{s}_x, \bar{s}_y$.

The S20_MK2 priority is

$$\pi_{\mathrm{S20_MK2}}(Q) = \lambda(Q) + 20\, S(Q) + 2\, \mathrm{MK}(Q).$$

Example 1.12. For AK(3): $\lambda = 13$, $\bar{s}_x = 6/4$ and $\bar{s}_y = 7/4$, so $S = 1.5$ and $\mathrm{MK} = 3$. Hence $\pi_{\mathrm{S20_MK2}} = 13 + 30 + 6 = 49$.

The weights were selected on a 120-row training split of hard AC1M orbits and evaluated once on two disjoint 60-row holdouts. There S20_MK2 solves 54/120, 33/60 and 27/60 at $B = 10^3$, against 0 in each case for LENGTH.

Score of S20_MK2 (cap 24 unless stated). Each entry is S20_MK2 against LENGTH.

set, budget	S20_MK2	LENGTH	only S20_MK2	only LENGTH
MS640, $B = 10^3$	574	554	24	4
MS640, $B = 10^4$	626	606	20	0
bench60, $B = 10^3$	37	29		
bench60, $B = 10^4$	52	40	12	0
ladder200, $B = 10^4$	116	40		
AC19, $B = 10^4$, cap 48	72,519	71,936		

Cost. On the 606 MS640 rows solved by both at $B = 10^4$, S20_MK2 uses 143,722 pops in total against 236,053 for LENGTH, a ratio of 0.61. At $B = 10^6$ both solve all 640 rows, with a mean of 994.8 pops per row against 4,963.0, a ratio of 0.20. Its certificates are longer: the mean path length is 42.88 S-moves against 36.77.

1.2 Stable AC conjecture

Definition 1.13 (Stable AC moves)**.** Let $z \notin X$. In addition to AC1–AC3, allow:

$$\begin{aligned} &\text{(AC4)} \quad \langle X \mid R\rangle \mapsto \langle X \sqcup \{z\} \mid R,\, z\rangle; \\ &\text{(AC5)} \quad \langle X \sqcup \{z\} \mid R,\, z\rangle \mapsto \langle X \mid R\rangle, \qquad \text{if } z \text{ does not occur in } R. \end{aligned}$$

Write $P \sim_s Q$ if Q is obtained from P by finitely many moves AC1–AC5. P is *stably AC-trivial* if $P \sim_s T_n$, where n is its rank.

AC4 and AC5 replace M by $M \oplus (1)$ and back, so Lemma 1.3(2) persists. AC-triviality implies stable AC-triviality.

Conjecture 1.14 (Stable Andrews–Curtis)**.** *Every balanced presentation of the trivial group is stably AC-trivial.*

It is open whether stable AC-triviality implies AC-triviality, and the stable conjecture is itself open.

Lemma 1.15 (Invariance under automorphisms)**.** *Let $\varphi \in \mathrm{Aut}(F(X))$ and let $\varphi(P)$ denote $\langle X \mid \varphi(r_1), \ldots, \varphi(r_n)\rangle$. Then P is AC-trivial if and only if $\varphi(P)$ is AC-trivial.*

Proof. φ commutes with every AC move, because

$$\varphi(r_i r_j) = \varphi(r_i)\varphi(r_j) \quad \text{and} \quad \varphi(uru^{-1}) = \varphi(u)\varphi(r)\varphi(u)^{-1}.$$

Hence a sequence $P \to T_n$ maps to a sequence

$$\varphi(P) \longrightarrow \langle X \mid \varphi(x_1), \ldots, \varphi(x_n)\rangle.$$

The tuple $(\varphi(x_i))_i$ is a basis of $F(X)$. By Nielsen's theorem it is carried to $(x_i)_i$ by elementary Nielsen transformations (inversion of an entry, multiplication of one entry by another, permutation), which are AC1, AC2 and relator transpositions. Apply the same argument to φ^{-1} for the converse. □

1.2.1 3-generators search

Definition 1.16 (Stabilised start)**.** For a balanced $P = \langle X \mid R\rangle$, words $w_1, \ldots, w_k \in F(X)$ and new generators $z_1, \ldots, z_k \notin X$, let

$$P_{w_1,\ldots,w_k} = \langle X \sqcup \{z_1, \ldots, z_k\} \mid R,\ z_1^{-1}w_1,\ \ldots,\ z_k^{-1}w_k\rangle.$$

For $k = 1$ and $|X| = 2$, $P_w = \langle x, y, z \mid r_1, r_2, z^{-1}w\rangle$ is the *rank-3 start*. It presents G_P.

Proposition 1.17. *For all $w_1, \ldots, w_k \in F(X)$,*

$$P \text{ AC-trivial} \implies P_{w_1,\ldots,w_k} \text{ AC-trivial} \implies P \text{ stably AC-trivial.}$$

Proof. It suffices to treat $k = 1$ and iterate. Let $P^+ = \langle X, z \mid R, z\rangle$. Let $\varphi \in \mathrm{Aut}(F(X, z))$ fix X and send $z \mapsto w^{-1}z$; its inverse is $z \mapsto wz$. Since φ fixes R, $\varphi(P^+) = \langle X, z \mid R, w^{-1}z\rangle$, which is P_w after one AC1 move. By Lemma 1.15, P_w is AC-trivial if and only if P^+ is.

- If $P \sim_{AC} T_n$, the same moves give $P^+ \sim_{AC} T_{n+1}$.

- If $P^+ \sim_{AC} T_{n+1}$, then $P \xrightarrow{\text{AC4}} P^+ \sim_{AC} T_{n+1} \xrightarrow{\text{AC5}} T_n$.

□

A solution of P_w therefore certifies stable AC-triviality of P and nothing stronger.

Remark 1.18. 1. For $w = 1$, the start $P_1 = \langle X, z \mid R, z^{-1} \rangle$ has its extra relator already of length 1. On every MS640 row, at both budgets, the search on P_1 coincides with the rank-2 search: same outcome, same number of pops, same path length.

2. $P_{w^{-1}}$ is the image of P_w under $z \mapsto z^{-1}$ up to AC1, so the two starts are AC-trivial together by Lemma 1.15. The greedy's tie-break is not invariant under this renaming. For $|w| \leq 6$, 80.3% of the 465,920 pairs $(P_w, P_{w^{-1}})$ agree in outcome and pops, and 0.6% differ in outcome.

Definition 1.19 (Word family)**.** $\mathcal{W}_\ell$ consists of one word from each pair $\{w, w^{-1}\}$ of freely reduced words of length $\ell \geq 1$ over $\{x, y\}^{\pm 1}$. There are $4 \cdot 3^{\ell-1}$ such words, and $w \neq w^{-1}$ for $w \neq 1$ because F_2 is torsion-free. Hence

$$|\mathcal{W}_\ell| = 2 \cdot 3^{\ell-1}, \qquad \left|\mathcal{W}_{\leq L}\right| = \sum_{\ell=1}^{L} 2 \cdot 3^{\ell-1} = 3^L - 1.$$

Definition 1.20 (Measures)**.** Fix a benchmark D, a budget B and the LENGTH greedy at cap 24. For $P \in D$, let n_P be the number of pops of the rank-2 search on P and $n_{P,w}$ the number of pops of the search on P_w. Define

$$\begin{aligned} \mathcal{S}_B &= \{P \in D : \text{ the rank-2 search solves } P\}, \\ \mathcal{F}_B(w) &= \{P \in D : \text{ the search solves } P_w\}, \\ \mathcal{O}_B(\mathcal{W}) &= \textstyle\bigcup_{w \in \mathcal{W}} \mathcal{F}_B(w), \\ \text{gained} &= \mathcal{O}_B \setminus \mathcal{S}_B, \qquad \text{lost} = \mathcal{S}_B \setminus \mathcal{O}_B. \end{aligned}$$

$|\mathcal{O}_B(\mathcal{W})|$, the *oracle* score, counts the presentations for which some word of $\mathcal{W}$ succeeds. It answers an existence question and is not a procedure; its cost is the sum over all $w \in \mathcal{W}$. $\max_w |\mathcal{F}_B(w)|$ is the score of the best single word for the whole benchmark, chosen with hindsight.

Statistics over a random word are taken as follows. Draw P uniformly from the benchmark, then w uniformly from the family. Each presentation therefore has equal weight. The *node ratio* is $n_{P,w}/n_P$, computed over the pairs in which both searches succeed.

Score at $B = 10^3$, all 59,048 words of length 1 to 10 (37,790,720 searches; the rank-2 search solves $|\mathcal{S}| = 554$). The columns are:

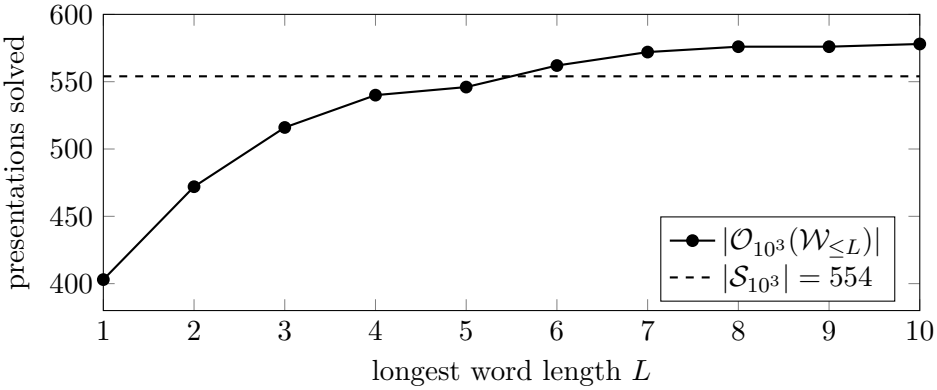
(a) the percentage of random pairs (P, w), $w \in \mathcal{W}_\ell$, that are solved;

(b)–(d) restricted to the 554 presentations of $\mathcal{S}$: the percentage of pairs in which the rank-3 search needs fewer pops than the rank-2 search, needs more pops, or fails within the budget. The remaining category (equal pops) never exceeds 0.06%;

(e), (f) the median and the geometric mean of the node ratio $n_{P,w}/n_P$, over pairs where both searches succeed;

(g)–(i) cumulative over $\mathcal{W}_{\leq \ell}$: the oracle score $|\mathcal{O}(\mathcal{W}_{\leq \ell})|$, the number gained and the number lost.

ℓ	$\lvert\mathcal{W}_\ell\rvert$	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
1	2	60.1	0.00	69.4	30.6	1.20	1.92	403	0	151
2	6	68.4	1.29	77.7	21.0	1.40	2.15	472	0	82
3	18	69.8	0.65	80.0	19.4	1.60	2.39	516	0	38
4	54	68.6	1.24	77.9	20.8	1.60	2.31	540	4	18
5	162	67.3	0.97	76.8	22.2	1.71	2.37	546	8	16
6	486	66.8	0.75	76.4	22.8	1.75	2.43	562	16	8
7	1,458	66.5	0.70	76.1	23.2	1.80	2.41	572	20	2
8	4,374	66.2	0.65	75.8	23.5	1.89	2.47	576	22	0
9	13,122	65.9	0.61	75.5	23.9	2.00	2.52	576	22	0
10	39,366	65.7	0.57	75.3	24.1	2.00	2.56	578	24	0
≤ 10	59,048	65.8	0.59	75.4	24.0	2.00	2.53	578	24	0



The measurements give the following statements for $B = 10^3$.

1. A uniformly random pair (P, w) with $|w| \leq 10$ is solved with probability 65.8%, against $554/640 = 86.6\%$ for the rank-2 search. On the 554 presentations of $\mathcal{S}$, the rank-3 search needs fewer pops than the rank-2 search in 0.59% of the pairs, more pops in 75.4%, and fails within the budget in 24.0%.

2. When both searches succeed, the rank-3 search pops a median of 2.00 times as many states as the rank-2 search. The geometric mean of the ratio is 2.53 and the mean excess is 110.6 pops.

3. Charging a failure the full budget, the expected number of pops of a random rank-3 start is 2.06 times that of the rank-2 search over MS640, and 4.56 times over the 554 presentations of $\mathcal{S}$.

4. The oracle exceeds the rank-2 score only for $L \geq 6$. It reaches 578 (24 gained, 0 lost) at $L = 10$, while $|\mathcal{W}_{\leq L}|$ grows by a factor of 3 per unit of L.

5. $\max_w |\mathcal{F}(w)| = 528$, attained by $w =$ `yXYYYxyyy`; no word attains 554. A word selected on one half of MS640 and scored on the other half solves 258 and 264 of 320, against 281 and 273 for the rank-2 search.

6. All 24 gained presentations lie in $\mathcal{S}_{B'}$ for some $B' \leq 26{,}838$ (20 of them for $B' \leq 10^4$). Enumerating $\mathcal{W}_{\leq 10}$ in shortlex order until a word succeeds costs at least 41,711 pops on each of them.

Score at $B = 10^4$, words of length 1 to 7 (2,186 words, 1,399,040 searches; $|\mathcal{S}| = 606$).

	$B = 10^3$	$B = 10^4$
rank-2 score $\lvert\mathcal{S}_B\rvert$	554	606
oracle $\lvert\mathcal{O}_B(\mathcal{W}_{\le 7})\rvert$	572	608
gained / lost	20 / 2	6 / 4
best single word $\max_w \lvert\mathcal{F}_B(w)\rvert$	512	572
random pair solved	66.5%	80.9%
on $\mathcal{S}_B$: fewer / more / unsolved	0.70 / 76.1 / 23.2%	0.85 / 84.5 / 14.7%
median node ratio (both solve)	1.80	2.00
expected pops relative to rank 2 (all / on $\mathcal{S}_B$)	–	2.74 / 5.24

Increasing the budget by a factor of 10 reduces the oracle's margin over the rank-2 search from 18 presentations to 2. With a fixed word for every row at $B = 5 \cdot 10^5$, the choices $w = r_1, r_2, x, y$ solve 619, 602, 540 and 523 presentations, against 638 for the rank-2 search.

Validity.

- *Cap.* The cap discarded a child in 2,331,711 of the 40,098,032 searches of the study (ranks 3 to 5, both budgets). In none of them does the condition of Remark 1.10(2) fail, so every score holds for every cap $c \ge 24$.
- *Relabelling.* MS640 consists of 319 presentations up to the renaming of x, y, each listed two or four times. Between two copies, the node count of a rank-3 search differs by more than 14–21% in one pair in ten, and the outcome differs in at most 1.8% of pairs. Counting each class once, the rank-2 search solves 276/319 and the oracle 288/319 (12 gained, 0 lost).
- *Verification.* Every stored path (more than 120,000) was replayed by a separate implementation, and 2,570 stored results were recomputed with a reference solver, with 0 mismatches. Seven independent checking programs, written from a prose specification, reported 0 mismatches.
- *Cost.* 1,329 core-hours (0.13 core-seconds per search) at $B = 10^3$, $L \le 10$; 354 core-hours at $B = 10^4$, $L \le 7$.

1.2.2 n-generators search

(a) Several stabilising words. Take $k = 2$ and $k = 3$ in Definition 1.16, with every unordered k-tuple of words, repetitions allowed, at $B = 10^3$. Each row is compared with rank 3 on the same word pool.

	rank 3, $\mathcal{W}_{\le 3}$	rank 4, $\mathcal{W}^2_{\le 3}$	rank 3, $\mathcal{W}_{\le 2}$	rank 5, $\mathcal{W}^3_{\le 2}$
starts per presentation	26	351	8	120
random start solved	68.7%	59.1%	66.3%	53.6%
on $\mathcal{S}$: fewer pops	0.75%	0.18%	0.97%	0.05%
on $\mathcal{S}$: unsolved	20.6%	31.7%	23.4%	38.1%
geometric-mean node ratio	2.31	3.51	2.16	3.59
oracle (rank 2: 554)	516	506	472	416
gained / lost	0 / 38	0 / 48	0 / 82	0 / 138
best single start	478	454	449	376

On an equal word pool, each additional generator lowers the oracle score, the success probability of a random start and the score of the best single start. It also raises the node ratio.

(b) Transformation to short relators.

Definition 1.21 (Definition step and triangulation)**.** Let $d = ab$ be a cyclic digram occurring in R. The *definition* $h := d$ adds a generator h and the relator $d^{-1}h$, and rewrites each relator by replacing occurrences of d by h (and d^{-1} by h^{-1}) wherever this shortens it. *Triangulation* applies definitions while some relator has length greater than 3, each time choosing the digram with the most cyclic occurrences.

Lemma 1.22. *Triangulation terminates, and its output $\tau(P)$ satisfies*

$$\tau(P) \text{ AC-trivial} \implies P \text{ stably AC-trivial.}$$

Proof. Each definition shortens a relator of length greater than 3, so the excess $\sum_i \max(0, |r_i| - 3)$ strictly decreases. Each definition is an AC4 move followed by the automorphism $h \mapsto d^{-1}h$ (as in Proposition 1.17) and AC moves. □

The code lifts a solution of the root $\tau(P)$ to elementary stable moves on P and replays them. On MS640 the root has rank 4 to 11 (mean 6.99).

Score. At $B = 10^4$, cap 24, the search on $\tau(P)$ solves 298/640 of MS640 against 606 for the rank-2 search, and 8/60 of bench60 against 40. Every presentation solved through $\tau(P)$ is also solved at rank 2. The lifted certificates have a median of 690.5 elementary moves (maximum 2,230).

(c) Rank changes during the search. The hybrid solver interleaves definition steps (rank $n \to n + 1$) and elimination steps with the rank-2 search. An elimination solves for a generator occurring once in a relator and substitutes it ($n \to n - 1$). Its budget is 10^3 *work units*, which are not pops, so its scores are not comparable with the tables above. It solves 640/640 of MS640, 60/60 of bench60, 188/188 of ladder200 and 72,779/72,779 of AC19. Of the AC19 certificates, 188 use definition or elimination steps and are therefore stable certificates.

1.2.3 Change of Variables (CoV)

Definition 1.23 (Change of variables)**.** Let $P = \langle x, y \mid r_1, r_2 \rangle$ and let $w \in F(x, y)$.

1. Stabilise to $P_w = \langle x, y, z \mid r_1, r_2, z^{-1}w \rangle$.
2. Replace non-overlapping occurrences of $w^{\pm 1}$ in r_1, r_2 by $z^{\pm 1}$ and freely reduce.
3. Choose a relator in which a generator $g \in \{x, y\}$ occurs exactly once together with z. Write it cyclically as $g^{\epsilon}s$, so that $g = s^{-\epsilon}$ in the group.
4. Substitute $g = s^{-\epsilon}$ into the other two relators and delete the chosen relator.
5. Rename the remaining two generators as x, y.

The result is a rank-2 presentation P', the *image*, and P' is searched with the unchanged rank-2 greedy.

Lemma 1.24. *If the image P' is AC-trivial, then P is stably AC-trivial. If, in addition, $P' = \varphi(P)$ for some $\varphi \in \mathrm{Aut}(F_2)$ up to AC moves, then P is AC-trivial.*

Proof. Step 1 is Proposition 1.17. Steps 2–4 are AC moves on P_w followed by AC5 after an automorphism that isolates g. The second statement is Lemma 1.15. □

The default family takes w among the cyclic subwords of length at least 2 of the relators. It discards any w whose substitution leaves a relator of length at most 2. The *one-shot policy* searches P first and then the images in a fixed order, with 10^3 pops per image and at most 10^4 pops in total per presentation. The exhaustive family keeps every proper cyclic subword and every set of non-overlapping occurrences, with one image per class under relator order, rotation, inversion and the 8 signed permutations: 114,042 images over MS640.

Score of the one-shot policy (10^3 pops per image, 10^4 per presentation) against the rank-2 greedy at $B = 10^4$:

set	CoV	of which stable certificates	LENGTH greedy
MS640	584	30	606
bench60	39	10	40
ladder200	92	48	40

At equal total pops the policy scores below the rank-2 greedy on MS640 and bench60, and above it on ladder200, whose rows are graded by the cost of the LENGTH greedy.

Score of the exhaustive oracle on MS640. Each of the 114,042 images is searched with the full budget.

budget per image	rank-2 $\lvert\mathcal{S}_B\rvert$	oracle	oracle − rank 2	mean pops per presentation to search all images
10^2	455	555	100	5,328.4
10^3	554	604	50	–
10^4	606	626	20	164,758.7
10^5	634	640	6	657,198.0

At $B = 10^4$, a uniformly random image of a presentation needs fewer pops than the rank-2 search in 33.0% of the pairs, the same number in 9.6%, more in 48.3%, and fails in 9.2%. The 20 presentations gained at 10^4 coincide with the 20 presentations that S20_MK2 solves and LENGTH does not.

Cap. Images were searched at cap max(24, longest relator+16) (between 24 and 91), the rank-2 search at cap 24. The comparison therefore favours CoV.

CoV against the rank-3 start on the same words. With w ranging over the proper cyclic subwords of the relators, the oracle of the rank-3 start solves 544 at $B = 10^3$ and 592 at $B = 10^4$, against 604 and 626 for the CoV images built from the same words. Every presentation solved by the rank-3 start is also solved by CoV. The two constructions differ only in the elimination of a generator after the substitution, which therefore accounts for a difference of 60 presentations at $B = 10^3$ and 34 at $B = 10^4$.

1.3 Automorphisms

Definition 1.25 (Whitehead automorphisms and canonical form)**.** The Whitehead automorphisms of F_2 are the 8 signed permutations of $\{x, y\}$ together with 12 automorphisms of the second kind, such as $x \mapsto xy$. *Peak reduction* applies Whitehead automorphisms that lower λ until none does. The *canonical form* of P is the lexicographic minimum of the set of presentations of minimal total length in its orbit. It is a complete invariant of the $\mathrm{Aut}(F_2)$-orbit of P modulo the state equivalence.

Proposition 1.26 (Transport)**.** *Let $\varphi \in \mathrm{Aut}(F_2)$ and let γ be a sequence of AC moves from $\varphi(P)$ to T_2. Then φ^{-1} applied move by move to γ, followed by the Nielsen moves of a factorisation of φ into Whitehead automorphisms, is a sequence of AC moves from P to T_2.*

Proof. This is the proof of Lemma 1.15, applied with φ^{-1}. An S-move is mapped to the S-move whose conjugators are the images of the original conjugators. □

The *aut* method searches the canonical form of P with S20_MK2 and transports the solution back to P. Its certificates are AC certificates on P.

Score.

- *Orbit counts.* MS640 contains 113 orbits, the 156,762 extended AC19 presentations contain 72,779, and the 1,136,154 AC1M presentations contain 71,283.
- *The aut method* at $B = 10^4$, cap 24, solves 626/640 of MS640 (11 plain and 615 transported certificates) and 52/60 of bench60. These equal the S20_MK2 scores on the input presentations.
- *The canonical form need not be easier.* On the original presentations of the 40 hardest AC19 orbits, the LENGTH greedy solves 40/40 (median 5,720 pops, maximum 52,143) and S20_MK2 solves 18/18. Within 10^7 pops, none of their canonical representatives is solved (0/28 with LENGTH, 0/9 with S20_MK2). Minimisation shortens the input on 34 of the 40 presentations and raises the peak length on all 40 (median 29 to 49).
- *Transport.* All 58 transported certificates replay on the representatives, with every one of the 2,211 and 1,069 transported steps equivariant.